

Davenport–Schinzel Sequences: The Pigeonhole Threshold

1 Definition

Definition 1 (Order- s Davenport–Schinzel sequence). Let $n, s \geq 1$ and $[n] = \{1, \dots, n\}$. A sequence $U = (u_1, \dots, u_m)$ over $[n]$ is an order- s Davenport–Schinzel sequence if

1. $u_i \neq u_{i+1}$ for every i ; and
2. for every two distinct symbols a, b , the sequence U contains no alternating subsequence

$$a, b, a, b, \dots$$

of length $s + 2$.

We write $U \in \text{DS}(n, s)$.

Definition 2 (Extremal function).

$$\lambda_s(n) = \max\{|U| : U \in \text{DS}(n, s)\}. \quad (1)$$

For all n and s , the pigeonhole argument gives

$$\lambda_s(n) \leq \binom{n}{2} (s + 1). \quad (2)$$

2 Current best result

This bound is asymptotically tight when

$$\frac{s}{\sqrt{n}} \rightarrow \infty. \quad (3)$$

In this regime,

$$\lambda_s(n) \sim \binom{n}{2} s. \quad (4)$$

3 Problem Formulation

Determine the smallest growth rate of $s = s(n)$ for which

$$\lambda_s(n) \sim \binom{n}{2} s. \quad (5)$$

In particular, determine whether this equivalence can hold for some $s = o(\sqrt{n})$.